

Implementation of the Varatharajan 2005 Ethylene Detonation Mechanism in Uintah’s ICE Component

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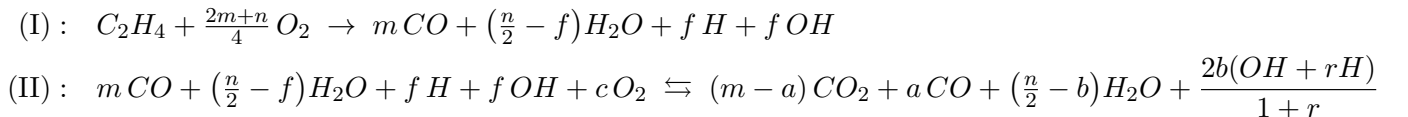
1 Introduction

The following model is the implementation of the two-step chemical kinetic mechanism for ethylene detonation proposed in *Two-step chemical-kinetic descriptions for hydrocarbon–oxygen–diluent ignition and detonation applications* by B. Varatharajan et al. (2005). The following document describes the steps and equations used to simulate the combustion of ethylene in air. The following assumptions and simplifications have been made:

1. ICE simulations are built on a single material, Air. Individual species are not known by ICE, only their representation as passive scalars. Therefore, phenomena like molecular diffusion and specific heats due to composition change are not modelled.
2. Species (C_2H_4 , O_2 , N_2 , CO_2 , CO , H_2O , H , OH) are tracked. Six passive scalars are tracked (C_2H_4 , O_2 , CO_2 , CO , H_2O , H). Y_{OH} is recovered from mass fraction closure.
3. Molar enthalpy data for heat release is calculated using NASA 7 polynomials. Coefficients are taken from the GRI-Mech 3.0 database and cover species [C_2H_4 , O_2 , CO_2 , CO , H_2O , H , OH].
4. This model modifies only the energy source term and the six passive scalar source terms. Source terms are computed as instantaneous rates multiplied by the timestep.
5. Mixture specific heat c_v and ratio of specific heats γ are not modified by this model. ICE computes these properties externally, treating the working fluid as ideal air ($O_2 : 3.76 N_2$) throughout the simulation.

2 Global Reactions

The mechanism consists of two global reaction steps derived in Varatharajan et al. for ethylene ($m = 2$, $n = 4$, $f = 2/3$). Step I is irreversible and describes fuel consumption producing CO , H_2O , and the radicals H and OH . Step II is reversible and describes CO oxidation and radical recombination approaching chemical equilibrium:



2.1 Stoichiometric Coefficients

The stoichiometric coefficient vectors $\boldsymbol{\nu}$ (products – reactants) for the 7 reactive species $[C_2H_4, O_2, CO_2, CO, H_2O, H, OH]$ are:

$$\boldsymbol{\nu}^I = \left[-1, -\frac{2m+n}{4}, 0, m, \frac{n}{2} - f, f, f \right]$$

The Step II coefficients depend on downstream equilibrium conditions through the parameters a, b, c, r , and s defined in Varatharajan et al.:

$$\begin{aligned} r &= \frac{[H]_\infty}{[OH]_\infty} & s &= \frac{k_{3,\infty}[H_2O]_\infty}{k_{4,\infty}} \\ a &= \frac{m}{1 + k_{5,\infty} / (r k_{6,\infty})} & b &= \frac{n(1+r)}{2(1+r+2\sqrt{rs})} \\ c &= \frac{m-a}{2} + \frac{b(1-r)}{2(1+r)} \end{aligned}$$

where subscript ∞ denotes evaluation at post-reaction equilibrium. The coefficients are:

$$\boldsymbol{\nu}^{II} = \left[0, -c, m-a, a-m, f-b, \frac{2br}{1+r} - f, \frac{2b}{1+r} - f \right]$$

These are applied to the 7 reactive species only; N_2 contributes no source term.

3 Reaction Rates

3.1 Elementary Reactions

Both global rates are expressed in terms of a small set of elementary reactions from Table 2 of Varatharajan et al. (2005). Units: mol, cm³, s, K, kJ/mol.

No.	Reaction	A	n	E_a	Efficiencies (Default 1.0)
(1)	$H + O_2 \rightarrow OH + O$	3.52e16	-0.70	71.4	
(2)	$H + O_2 + M \rightarrow HO_2 + M$	2.60e19	-1.20	0.0	$O_2 : 0.3; CO_2 : 1.5; CO : 0.75; H_2O : 7.0$
(3)	$H + OH + M \rightarrow H_2O + M$	2.20e22	-2.00	0.0	$CO : 1.9; CO_2 : 3.8; H_2O : 12.0$
(4)	$H_2O + M \rightarrow H + OH + M$	2.18e23	-1.93	499.0	$CO : 1.9; CO_2 : 3.8; H_2O : 12.0$
(10)	$C_2H_4 + O_2 \rightarrow C_2H_3 + HO_2$	4.22e13	0.00	241.0	
(11)	$C_2H_4 + H + M \rightleftharpoons C_2H_5 + M$	$k_0 = 1.90e35, k_\infty = 1.08e12$	-5.57, 0.45	21.1, 7.6	Troe: $F_c = 0.832 \exp(-T/1203)$
(12)	$C_2H_5 + O_2 \rightarrow C_2H_4 + HO_2$	2.00e12	0.00	20.9	
(13)	$C_2H_5 + M \rightarrow C_2H_4 + H + M$	$k_0 = 3.99e33, k_\infty = 1.11e10$	-4.99, 1.04	167.4, 153.8	Troe: $F_c = 0.832 \exp(-T/1203)$

3.2 Global Rate I (Fuel Consumption)

Step I is irreversible. Its rate is the sum of an initiation term (k_{10}) and a chain-branching term keyed to the $H + O_2$ elementary reaction:

$$\omega_I = k_{10}[C_2H_4][O_2] + q k_1[H][O_2] \mathcal{H}\{[C_2H_4]\}$$

where $\mathcal{H}\{\cdot\}$ is the Heaviside function that turns off branching once fuel is exhausted. The correction factor q accounts for the slowdown of branching via HO_2 formation at lower temperatures:

$$q = \frac{k_1}{k_1 + k_2[M_2] + t k_{11}}, \quad t = \frac{k_{12}[O_2]}{k_{12}[O_2] + k_{13}[M_{\text{sum}}]}$$

where $[M_2]$ is the chaperon-weighted third body term for reaction 2, $[M_{\text{sum}}]$ is the unweighted total molar concentration, and k_{11} , k_{13} are Troe falloff rates.

3.3 Global Rate II (CO Oxidation / Radical Recombination)

Step II is reversible and driven by the $H + OH$ three-body recombination and its reverse:

$$\omega_{\text{II}} = \frac{k_3[M_3][H][OH] - k_4[M_3][H_2O]}{f}$$

where $[M_3]$ is the chaperon-weighted third body term for reactions 3 and 4.

3.4 Third Body Terms

Three distinct third body concentrations appear:

- $[M_{\text{sum}}] = \sum_k [s_k]$ — unweighted total molar concentration (Troe reactions 11, 13)
- $[M_2] = \sum_k \alpha_k^{(2)} [s_k]$ — chaperon-weighted, reaction 2 ($H + O_2 + M$)
- $[M_3] = \sum_k \alpha_k^{(3)} [s_k]$ — chaperon-weighted, reactions 3 and 4 ($H + OH + M$)

3.5 Troe Falloff (Reactions 11 and 13)

Both pressure-dependent reactions use center factor $F_c = 0.832 \exp(-T/1203)$. The falloff procedure follows:

$$P_r = \frac{k_0 [M_{\text{sum}}]}{k_\infty}, \quad \log_{10}(F) = \frac{\log_{10}(F_c)}{1 + \left(\frac{\log_{10}(P_r)}{0.75 - 1.27 \log_{10}(F_c)} \right)^2}, \quad k = k_\infty \frac{P_r}{1 + P_r} F$$

4 Functions

4.1 arrhenius(dbl T, int idx, dbl M)

Purpose: Evaluate the forward rate constant for elementary reaction **idx**.

Inputs: Temperature (T), reaction index **idx** (1-based), total molar concentration M (used only for Troe reactions 11 and 13).

Steps:

1. if (**idx** == 11 || **idx** == 13):
 - Compute **k0**, **kinf** via Arrhenius for low/high pressure limits
 - double **Pr** = **k0** * **M** / **kinf**;
 - double **Fc** = 0.832 * exp(-T / 1203.0);
 - double **N** = 0.75 - 1.27 * log10(**Fc**);
 - double **logF** = log10(**Fc**) / (1 + pow(log10(**Pr**) / **N**, 2));
 - return **kinf** * **Pr** * pow(10.0, **logF**) / (1 + **Pr**);
2. Otherwise: int **i** = **idx** - 1; return **A**[**i**] * pow(T, **n**[**i**]) * exp(-**E**[**i**] / (R * T));

4.2 globalRates(dbl T, vec Y, dbl rho, int rate)

Purpose: Compute global rate ω_I (rate=1) or ω_{II} (rate=2) in mol/cm³·s.

Inputs: Temperature (T), 8-species mass fraction vector Y (including N_2), density in kg/m³ (rho), rate selector (rate).

Steps (rate = 1):

```
1. double rho_g = rho / 1000.0;
2. conc[j] = rho_g * Y[j] / MM_N2[j]; ← for all species
3. if (Y[O2] < massTol || Y[C2H4] < massTol) return 0.0;
4. double iM13 = accumulate(conc); double iM2 = inner_product(conc, chap2);
5. double k1 = arrhenius(T,1,0); double k2 = arrhenius(T,2,0); double k10 = arrhenius(T,10,0);
6. double k11 = arrhenius(T,11,iM13); double k12 = arrhenius(T,12,0); double k13
   = arrhenius(T,13,iM13);
7. if (Y[H] < massTol) return k10 * conc[C2H4] * conc[O2];
8. double t = k12*conc[O2] / (k12*conc[O2] + k13*iM13);
9. double q = k1 / (k1 + k2*iM2 + t*k11);
10. double heavy = (conc[C2H4] > 1e-12) ? 1.0 : 0.0;
11. return k10*conc[C2H4]*conc[O2] + q*k1*conc[H]*conc[O2]*heavy;
```

Steps (rate = 2):

```
1. if ((Y[H] < massTol || Y[OH] < massTol) && Y[H2O] < massTol) return 0.0;
2. double iM3 = inner_product(conc, chap3);
3. double k3 = arrhenius(T,3,0); double k4 = arrhenius(T,4,0);
4. return (k3*iM3*conc[H]*conc[OH] - k4*iM3*conc[H2O]) / f;
```

4.3 molarEnthalpy(dbl T)

Purpose: Compute molar enthalpies (J/mol) for the 7 reactive species [C_2H_4 , O_2 , CO_2 , CO , H_2O , H , OH] using NASA 7 polynomials.

Inputs: Temperature (T).

Steps:

```
1. for (int j = 0; j < 7; j++):
2. h = 1000*R*(a6[j] + a1[j]*T + a2[j]*pow(T,2) + a3[j]*pow(T,3) + a4[j]*pow(T,4)
   + a5[j]*pow(T,5));
3. h_species.push_back(h);
4. return h_species;
```

4.4 heatRelease(dbl T, dbl rho, vec Y, vec nu1, vec nu2)

Purpose: Compute the net volumetric heat release rate (W/m^3).

Inputs: Temperature (T), density (rho), 8-species mass fractions (Y), stoichiometric coefficient vectors (nu1, nu2, each of length 7).

Steps:

```

1. double omegaI = 1e6 * globalRates(T, Y, rho, 1); [mol/m3·s]
2. double omegaII = 1e6 * globalRates(T, Y, rho, 2);
3. vector<double> h = molarEnthalpy(T);
4. double qI = omegaI * inner_product(nu1, h);
5. double qII = omegaII * inner_product(nu2, h);
6. return -(qI + qII); [W/m3]

```

4.5 massSource(dbl T, dbl rho, vec Y, vec nu1, vec nu2)

Purpose: Compute species mass source terms ($\text{kg}/\text{m}^3\cdot\text{s}$) for the 7 reactive species.

Inputs: Temperature (T), density (rho), 8-species mass fractions (Y), stoichiometric coefficient vectors (nu1, nu2).

Steps:

```

1. double omegaI = globalRates(T, Y, rho, 1); [mol/cm3·s]
2. double omegaII = globalRates(T, Y, rho, 2);
3. for (int k = 0; k < MM.size(); k++):
4. tmp = omegaI * nu1[k] + omegaII * nu2[k];
5. tmp = MM[k] * tmp * 1e3; ← convert mol/cm3·s → kg/m3·s
6. S.push_back(tmp);
7. return S; (N2 not included in MM, so it receives no source term)

```

5 Implementation

Source terms are computed in `computeModelSources`, called once per timestep Δt . Reaction rates are evaluated at the cell state at the beginning of each step and applied in a single pass.

5.1 Step 1: Build the Cell State

For each cell retrieve from the data warehouse: Δt , temperature T , density ρ , and the 6 tracked mass fractions. The N_2 mass fraction is inserted at index 2. Y_{OH} is recovered from closure and appended last:

$$\mathbf{Y} = [Y_{C_2H_4}, Y_{O_2}, Y_{N_2}, Y_{CO_2}, Y_{CO}, Y_{H_2O}, Y_H, Y_{OH}], \quad Y_{OH} = \max\left(1 - \sum_{k \neq OH} Y_k, 0\right)$$

5.2 Step 2: Species Source Terms ($f = \text{massSource}$)

`massSource(T, rho, Y, nu1, nu2)` returns $\mathbf{S}$ (kg/m³·s) for the 7 reactive species. Only the first 6 (excluding *OH*) are written to the passive scalar source arrays:

```
Ysrc[k] += S[k] * delT / rho
```

5.3 Step 3: Energy Source Term ($f = \text{heatRelease}$)

`heatRelease(T, rho, Y, nu1, nu2)` returns the volumetric heat release rate $\dot{q}$ (W/m³). The energy source (J) added to the cell is:

```
eng_src += heatRelease(T, rho, Y, nu1, nu2) * cellVol * delT
```